Unit 6
Corrosion

Corrosion :- "Destruction of metal thⁿ unwanted chemical or electrochemical reacⁿ betⁿ the surface of metal & environment.

* Causes of corrosion :-
ores are stable with less energy state but pure metal is in a higher energy state.

* Consequence of corrosion :-
1) Loss of efficiency of the process
2) Life of equipment & machinery get reduced.
3) possibility of accidents & hazards.
4) Necessity for replacing the corroded equipments etc.

* Types of corrosion.
1) Dry corrosion / Direct chemical / Atmospheric
- It occures due to direct chemical reaction betⁿ the metal & the gases presence in the environment.
2) Wet corrosion / Electeochemical / Immerse
Corrosion of a metal by aqueous conducting medium, with the formation of anodic & cathodic areas is known as wet or electeochemical corrosion.

1) Dry corrosion -
~~other than at~~ when metal get directly contact with CO_2, O_2, H_2S, halogens, like Cl_2, Br_2, I_2, & SO_2 corrosion takes place.
Three types of corrosion - a) due to oxygen.
b) due to other gases
c) due to Liq. metals

a) Corrosion due to oxygen

corrosion takes place due to oxygen & formed metal oxides.

Mech^n:-

i) Adsorption of O_2 molecule on metal surface.
molecule of oxygen are adsorbed rapidly by the surface of the metal. It is physical phenomenon.

ii) Dissociation of oxygen molecule in to atom.
oxygen molecule get dissociate into oxygen atom.

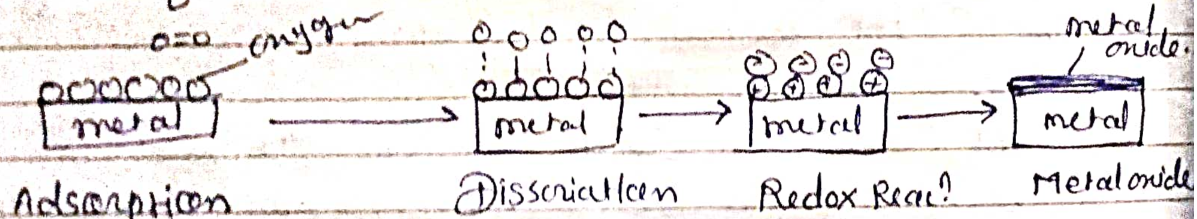
$$O_2 \xrightarrow{\text{Atomized}} 2O$$

iii) Redox reac^n
metal losses e^- & oxygen gain e^- so oxid^n & Red^n reac^n takes place.

IV) Formation of metal oxides.

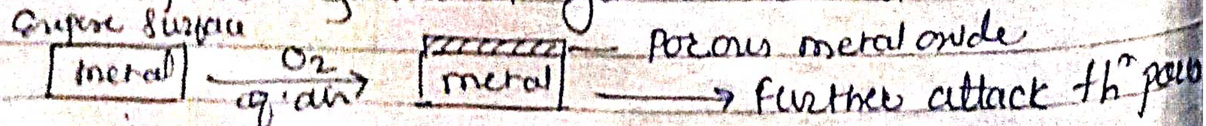
$$M^{++} + O^{--} \longrightarrow MO$$

when metal ion & oxygen ion comb^n together formed metal oxides.



* Nature of oxide film:-
i) Porous 2) Non-porous 3) unstable oxide film 4) volatile oxide film.

i) Porous oxide film (non protective)
If oxide film is porous then oxygen gas get penetrate th^r that & continuous corrosion takes place & metal get destructed.



eg - Li, K, Na, Mg.
1) $Na + O_2 \longrightarrow 2Na_2O$
2) $2Mg + O_2 \longrightarrow 2MgO$

ii) Non porous film:-
If oxide film is nonporous strong & well adher to metal ~~sur~~ surface. As a result corrosion is discontinued once oxide film is produced at the surface.
eg — Al, Cr, Cu, Zn,

1) $4Al + 3O_2 \longrightarrow 2Al_2O_3$
2) $4Cr + 3O_2 \longrightarrow 2Cr_2O_3$
3) $2Cu + O_2 \longrightarrow 2CuO$

iii) Unstable oxide film:-
When oxide film is unstable & decomposes back into metal & oxygen then there is no corrosion takes place. eg — Ag, Au, Pt.

1) $4Ag + O_2 \rightleftharpoons 2Ag_2O$
2) $4Au + 3O_2 \rightleftharpoons 2Au_2O_3$
3) $Pt + O_2 \rightleftharpoons PtO$

v) Volatile oxide film:-
If oxide film is volatile or get vaporizes as soon as formed. The underlaying metal surface is exposed for further attack of oxygen. & corrosion takes place continuous. eg Mo.
$Mo + O_2 \longrightarrow MoO_2 \uparrow$ Molybdenum.

* Pilling - Bedworth Rule:-
This rule gives the idea of ~~a~~ nature of oxide film. They classified — 1) metal forming protective oxide.
2) metal forming nonprotective oxide.

$$\text{Pilling-Bedworth Ration (PBR)} = \frac{\text{vol}^m \text{ of metal oxide formed}}{\text{vol}^m \text{ of metal atom consumed.}}$$

1) An oxide is protective or non porous if the oxide is least as greater than volm of metal.
2) If the ~~vo~~ volm of the oxide is less than volm of metal then it is porous & non-protective.

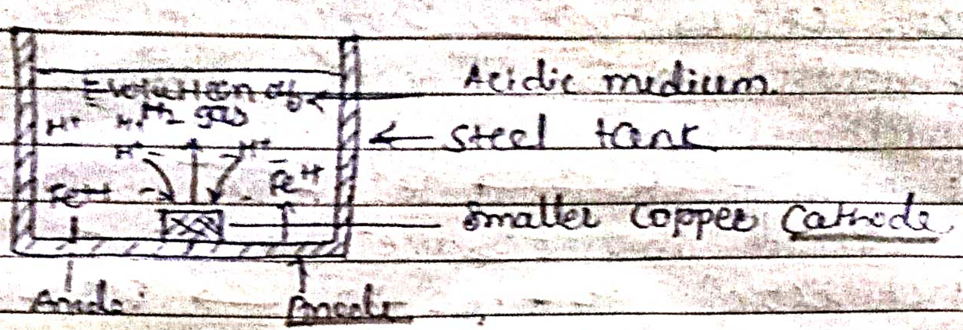
* Corrosion due to other gases :-
→ The Corrosion depends on chemical affinity betⁿ the metal & the gases.
i) Halogen / Chlorine :
When chlorine attack on metal formed metal chloride & that film is porous film
$2Fe + 3Cl_2 \longrightarrow 2FeCl_3 \downarrow$
ii) Corrosion by gases containing sulphur :-
SO_2, SO_3, H_2S attack on metal & causes corrosion.
eg $Fe + H_2S \longrightarrow FeS \downarrow + H_2 \uparrow$
iii) Hydrogen embrittlement (Decarburization)
when Hydrogen attack on metals' occures hydrogen embrittlement.
At high temp Hydrogen penetrate into steel & react with carbon to formed CH_4, i.e at higher temp. removal of carbon takes place from metal by the action of hydrogen.
$C + 4H \xrightarrow[\text{temp}]{\text{high}} CH_4$
iv) CO & CO_2
CO_2 is acidic in nature react with metals
$Fe + CO_2 \longrightarrow FeO + CO$.

* Wet / Electrochemical / Immersed Corrosion.
Corrosion of a metal by aqueous conducting medium with the formation of anodic & cathodic area is called wet corrosion.
* Mechanism of wet corrosion

Hydrogen evolution mech. | oxygen absorption mech.

a) Hydrogen evolution mech :-
This type of corrosion occurs in acidic environment like industrial waste solⁿ
eg — The piece of Cu & steel tank in contact

with each other in presence of acid electrolyte gives rise to an electrochemical cell.



In this cell steel act as anode & copper act as cathode.

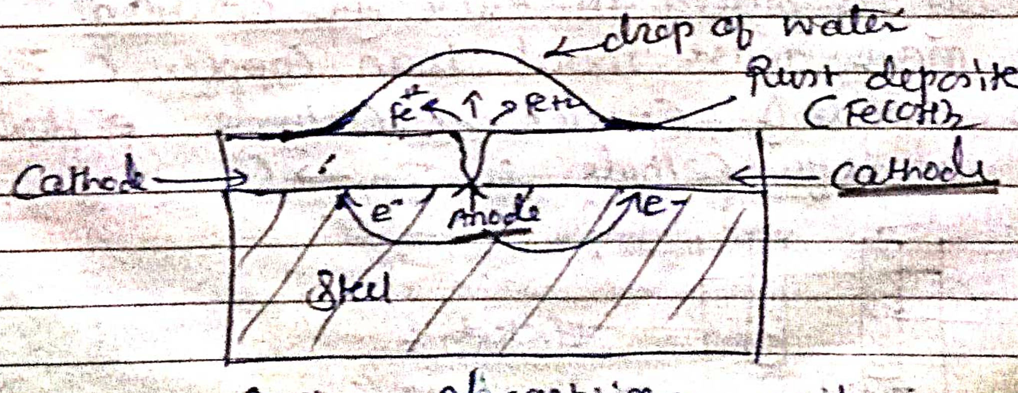
At Anode (steel tank) — $Fe \rightarrow Fe^{+2} + 2e^-$ (oxidⁿ)

At Cathode (copper) $2H^+ + 2e^- \rightarrow H_2$ (Redⁿ)

Net Reacⁿ $Fe + 2H^+ \longrightarrow Fe^{+2} + H_2 \uparrow$

- At cathode — Hydrogen ion from acidic medium takes up the free e^- & formed Hydrogen gas in the formed of bubbles.
- ~~At~~ In Hydrogen evolution mech of corrosion anode have larger area than cathode.

e) Oxygen absorption mechⁿ:—
Corrosion occures due to neutral aqueous solⁿ of electrolyte like NaCl.
eg- A steel ~~tank~~ plate laying on the ground & exposed to atmosphere formed oxide layer. The water drop collect in crack on the oxide film present on the surface of the steel.



Oxygen Absorption mech

At Anode — steel goes in to sol^n & formed Fe^{++}
At cathode - e^- flow from anode to cathode i.e from cracks to the surface of steel plate & dissolved oxygen present in electrolyte forming hydroxyl ion.
At Anode - $Fe \longrightarrow Fe^{+2} + 2e^-$ oxid^n
At cathode $H_2O + \frac{1}{2}O_2 + 2e^- \longrightarrow 2OH^-$ Red^n
net reac^n $Fe + H_2O + \frac{1}{2}O_2 \longrightarrow Fe^{+2} + 2OH^-$

$$\downarrow$$

$Fe(OH)_2 \downarrow$
(Ferrous hydroxide)

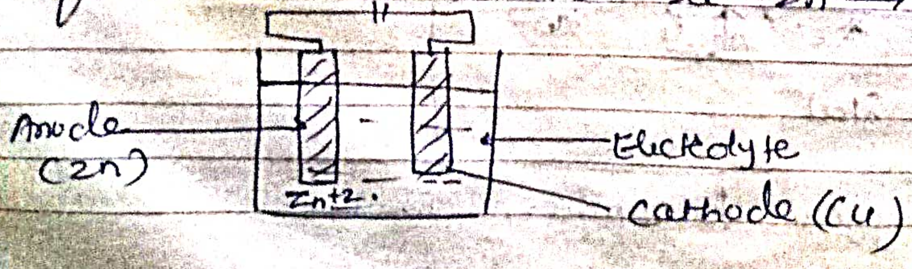
When Ferrous hydroxide again contact with oxygen & water formed ferric hydroxide.

$4Fe(OH)_2 + 2H_2O + O_2 \longrightarrow 4Fe(OH)_3 \downarrow$
(Ferric hydroxide)

In oxygen absorption reac^n mech. of corrosion, cathodic area is larger than anode.

* Types of wet corrosion :-
  1) Galvanic corrosion
  2) concentration / differential corrosion.

1) Galvanic corrosion :-
when two dissimilar metals are electrically connected & exposed to an electrolyte the metal higher the electrochemical series undergoes corrosion.
eg - when two metals zinc & copper are connected electrically & placed in electrolyte solution. Zn is more active metal than copper so Zn act as anode & Cu act as cathode. e^- flow from anode to cathode. i.e Zn → Cu.

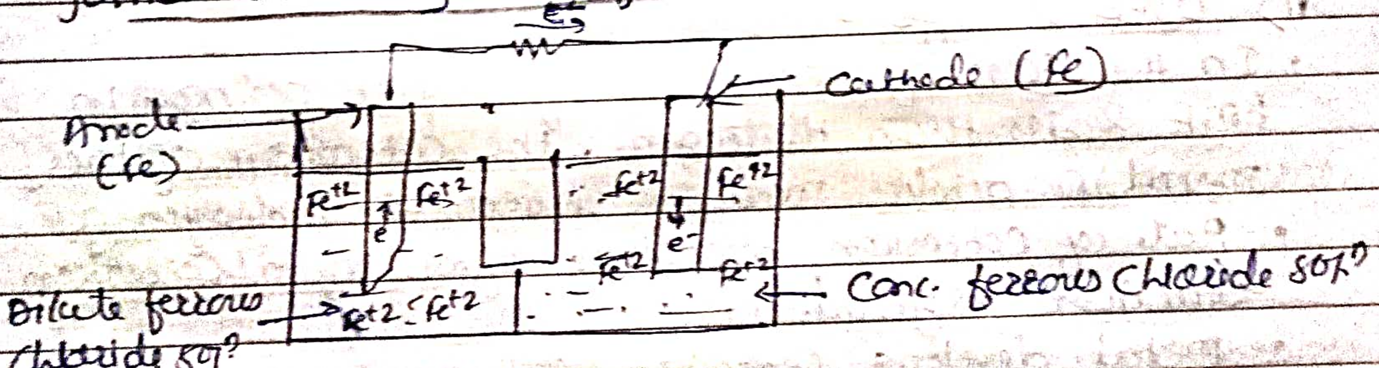


eg – Rusting of fencing wire under joints
~~Steel~~ Steel pipe joined by iron nails

2) Concentration cell corrosion:-
when a single piece of metal is exposed to an electrolyte of varying conc. or varying aeration is called conc cell corrosion.

a) Metal ion conc cell corrosion
b) Oxygen conc. cell corrosion

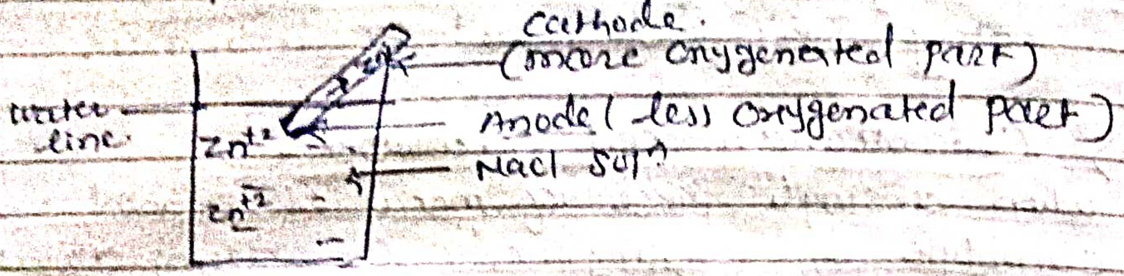
a) Metal-ion concentration cell corrosion:-
- A difference in the ~~ele~~ metal ion conc. of the electrolyte formed potential difference of two electrodes. Electrode potential of metal in dil. sol^n is more -ve than ~~the~~ in a conc. sol^n
- When two electrodes of same metal immersed into diff. conc solutions (concentrations) of same electrolyte are joined externally they formed metal ion conc. cell



- Iron dipped in low concentration solution act as Anode & Iron dipped in more conc. sol^n act as Cathode
- This corrosion occures in chemical plant.

2. Oxygen conc. cell corrosion:
- When two parts of the same metal are contact in different aeration cond^n the potential difference get developed.
- Less aerated part on the metal is more -ve act as anode & more aerated part act as cathode

$M \longrightarrow M^{+n} + ne^{-1}$



* Factors Influencing Rate of Corrosion:
1) Nature of metal
2) Nature of environment.

1) Nature of metal.

i) Electrode potential (oxid^n potential)
corrosion of metal depends on electrochemical series & galvanic series. metal placed higher the galvanic series become anodic so faster will be the corrosion takes place & other metal get protected.

ii) over voltage:-
- In the case of metal having higher position in EMF series than Hydrogen, the corrosion of the metal in acidic medium liberates hydrogen gas.
- Rate of corrosion decreases after initial corrosion reaction begined.
- metal develops corrosion resistance to some extent.
- To restored position of metal extra voltage is applied.

iii) Relative areas of anode & cathode:
- If the ratio of cathodic area to anodic area is more then rate of corrosion is increased.
- If anodic area es smaller current density is more & more demands of e^- from cathodic area. causes more corrosion.

iv) Purity of metal:-
The impurities present in corrosion causes

galvanic ~~act~~ cell action under environment & anodic part get corroded.
eg — Impurities like Pb, Fe in zinc lead formed electrochemical cells, corrosion of zinc takes place due to impurities or other metals.

v) Physical State of metal.
- rate of corrosion depends on physical state of the metal like grain size, stress, crystal structure.
- If smaller the grain size more solubility is there & corrosion takes place. Under stress even pure metal act as anode & corrosion takes place.

vi) Nature of oxide film:
- If the oxide film formed on metal surface is protective, nonporous, strongly adhered then neither dry or wet corrosion takes place.
- If the oxide film formed on metal surface is nonprotective, porous, loosely ~~d~~ adhered then corrosion takes place.

2 Nature of environment :-

i) Temperature :-
- The rate of dry corrosion is faster at high temp. at high temp. metal get activated.
- As per Neernst eqn for electrode potential, the potential of a metal is higher at high temperature hence rate of corrosion is faster at high temp.

ii) Moisture (Humidity) :-
- Corrosion of metal is faster in humid atmosphere than in dry atmosphere because moisture act as a solvent for atmospheric gases like O_2, CO_2, etc. & formed galvanic cell.
- Moisture reacts with the metal or metal oxide & causes electrochemical corrosion.

iii) pH :-
In Acidic condition corrosion rate increases

Faster. Acidic media are more corrosive than alkaline & neutral media. metal do not undergoes corrosion at pH greater than 10 due to formation of protective coating of hydrous oxide of iron. Bet^n pH 10 & 3 oxygen is essential for corrosion.

iv) Nature of ions present :-

- Presence of anions like silicate in the medium lead formation of insoluble reac^n product which inhibit further corrosion.
- Cl^- ions present in the medium it destroy the protective & passive surface film & rapid corrosion takes place.

v) Conductance of the corroding medium :-
The conductance of the medium is imp. in corrosion. Thus clayey & mineralized soil is more conductive than sandy soil so conduction rate is more in ~~sand~~ clayey soil.

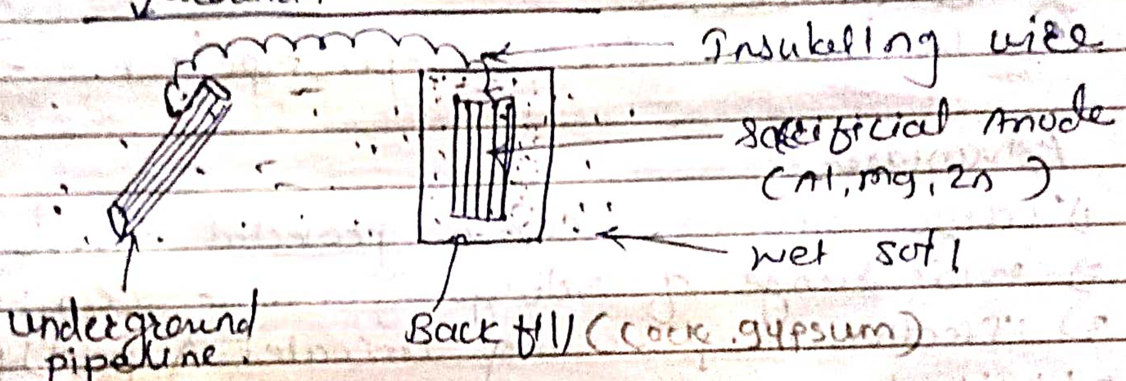
* * Corrosion control method *

- ① Cathodic protection
  - Sacrificial anode
  - Impressed current
- ② Anodic Protection
- ③ metallic coating
  - Hot dipping
    - Galvanising
    - Tinning
  - Electroplating
  - cementation
  - metal spraying
- ④ Chemical inorganic coating
- ⑤ Powder coating
- ⑥ Blocodising

① Cathodic protection :-
Principle :- Cathodic protection is to force the metal ~~bevel~~ behave like a cathode.
Two types of cathodic protection.
i) Using sacrificial anode.
ii) Using impressed current.

i) using Sacrificial Anode :

- In this tech. a more active metal is connected to the metal structure to be protected that all corrosion is concentrated at the more active metal & thus saving the metal from corrosion.
- This method is used for the protection of sea going vessels such as ships & boats. Sheets of Zn or Mg are hung around the hull of the ship.
- Zn or Mg becomes Anode & it get corroded. Fe act as a cathode & it protected.
- The corroded sacrificial anode is replaced by a fresh one when consumed completely.



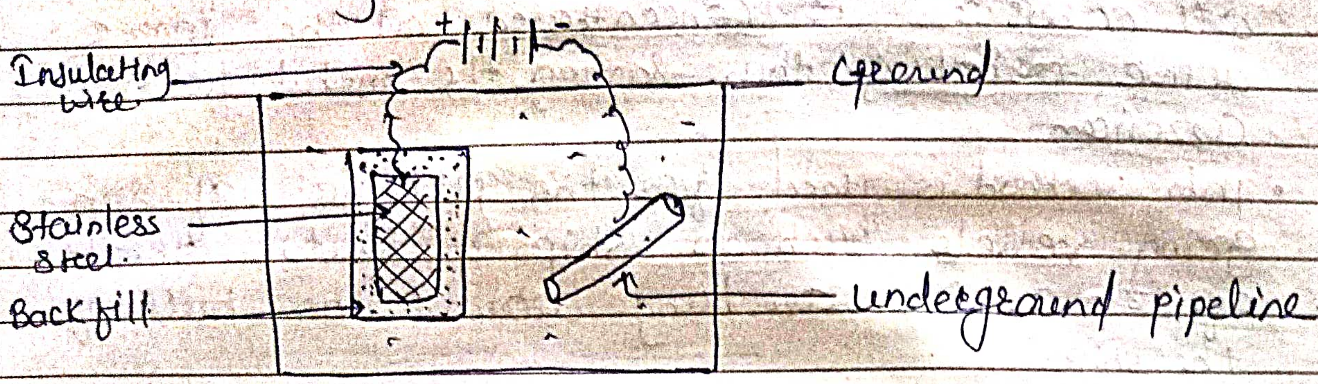
Applications :-

1. buried steel pipeline, 2. ship hull.
3. hot water tank 4. buried cables.

ii) Impressed current cathodic protection :

- In this method an impressed current is applied in an opposite direction to nullify the corrosive current & converting the corroding metal from anode to cathode.
- In this tech. direct current from a battery is used.
- Direct current is given to anode like stainless steel.
- -ve terminal connected to pipeline to be protected

Anode is kept in back fill (gypsum, coke breeze) to increase electrical contact with the surrounding soil.



Appl^n :-

① open water box cooler
② water tank.
③ Condenser,
④ Buried water or gas pipeline. etc.

Advantages :-

1) Large structure can be protected for long time.
2) Wide range of voltage & current can be used.
3) Effective in protecting uncoated & poorly coated.

* Limitations :-

① High capital investment & maintenance costs.
② Reqrd external power.

2) Anodic Protection :-

Principle :-

" metal or alloy having wider range of passivity voltage, is made anodic & the voltage in the passivity range is applied over it to control its corrosion even by strongly corroding media.

Construction & Working :-

① metallic tank is to be protected from corrosion connected to the potentiostat. (used for desired voltage & corrosion current.)

• Auxillary electrode act as cathode. metal tank to be protected act as a anode.
• If system goes out of the corrosion control any time, the corrosion current will be high & the electrical connections are removed immediately.

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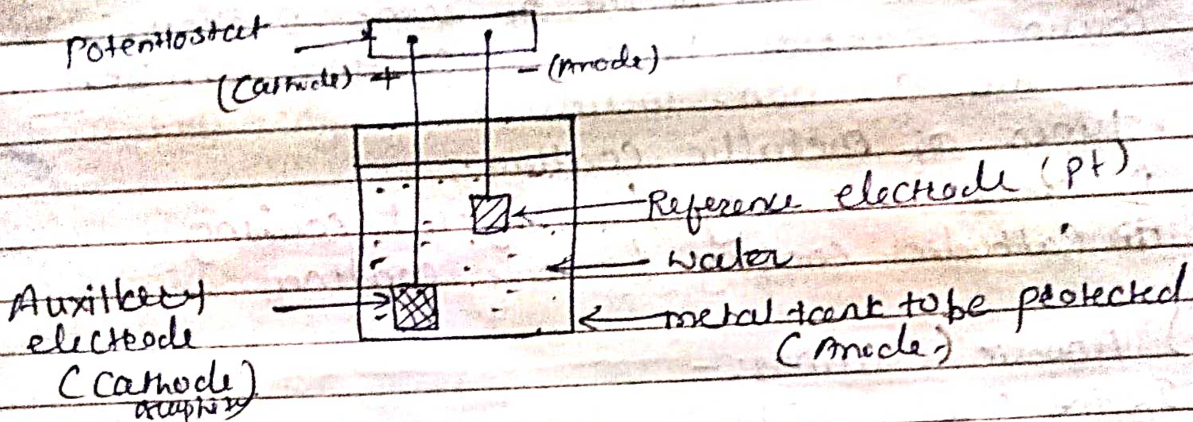

*Advantages :-
• Low operating cost.
• Applicable to highly corrosive media.
• Reliable to protect complex structure.
• Protection current gives idea about corrosion rate.

* Limitations :-
• High installation cost
• High starting current reqd.
• Appl^{n} only those metal which show active-passive nature.

* Appl^{n} :-
• Chemical reactor, tanks.
• Pipe carrying corrosive liq.
• Industrial water coolers.

cathodic protection	anodic protection
① Applicable to all metals	① Applicable only those metal show Active-passive nature.
② Installation cost low	② Installation cost more
③ operating cost high	③ operating cost low

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④ No source of power is reqrd. ② Reqrd source of power.

⑤ If system goes out of control slower the corrosion of metal. ③ If system goes out of control corrosion takes place faster.

3.) Metallic coating:-

metal can be protected by some protective coating material. Protective coatings can be metallic or non-metallic.

* Types of metallic coating:-

i) Anodic coating / sacrificial coating

ii) Cathodic coating / noble coating.

i) Anodic Coatings:-

If the coating metal is higher placed in galvanic series than the base metal then the coating is called anodic coating. eg Zn, Al, Cr coated on steel.

eg - In case of galvanised steel (steel coated with zinc) the coating metal zinc is more anodic to iron (steel) so it get corrode & steel get protected. Thus zinc coating protect iron it sacrificially, hence it is also called as sacrificial coating.

ii) Cathodic Coating / Noble coating:-

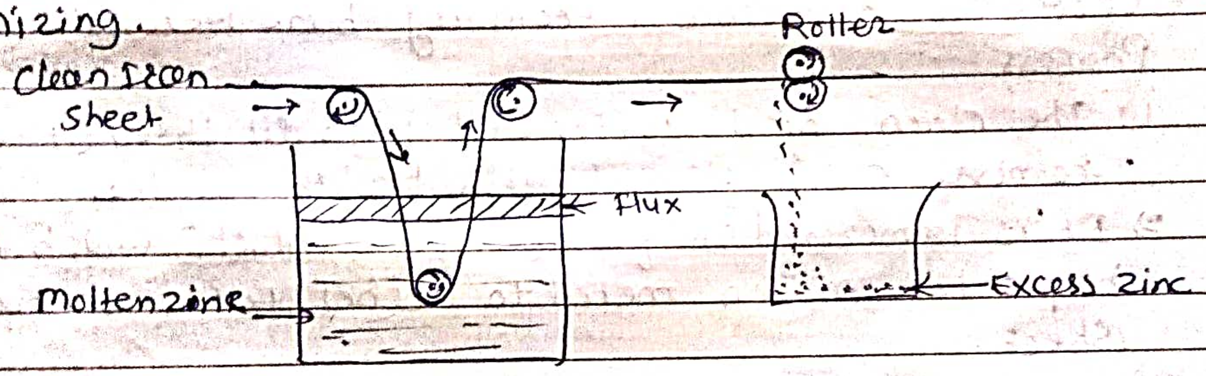
If the coating metal is placed in galvanic series than the base metal then the coating is called as cathodic coating - eg - Coating of tin on steel.

eg - Iron is base metal, & coating of tin will be cathodic coating. Cathodic coating provides effective protection to the base metal when it is free from pores & cracks. If coating is rupture then corrosion takes place fast.

* Methods of applying metallic coating:-
① Hot dipping ② Metal cladeling
③ Cementation ④ Electroplating

1) Hot dipping :-
Hot dipping is used for producing a coating of low melting metals such as Zn (419°c) or tin (232°c) on metal like steel, copper etc.
Types of Hot dipping
i) Galvanizing ii) Tinning.

i) Galvanizing :- (Zn)
The process of coating of Iron or steel (base metal) with coat of Zn (zinc) by hot dipping to prevent base metal corrosion is called galvanizing.



Galvanizing.

Process —
1) the metal articles is cleaned & degreased by using organic solvent
2) It is treated with dil. H_2SO_4 (pickling) for 15-20 min. at 70-90°c to remove any rust
3) It is washed with water & dried.
4) The metal article is then dipped in bath of molten Zn at 425°c to 430°c
5) The surface of bath is covered with flux ($NH_4Cl + ZnCl_2$)
Flux cleans the metal surface before coating for better adhesion & prevent the oxidation.
6) It is then passed thr pair of hot roller to

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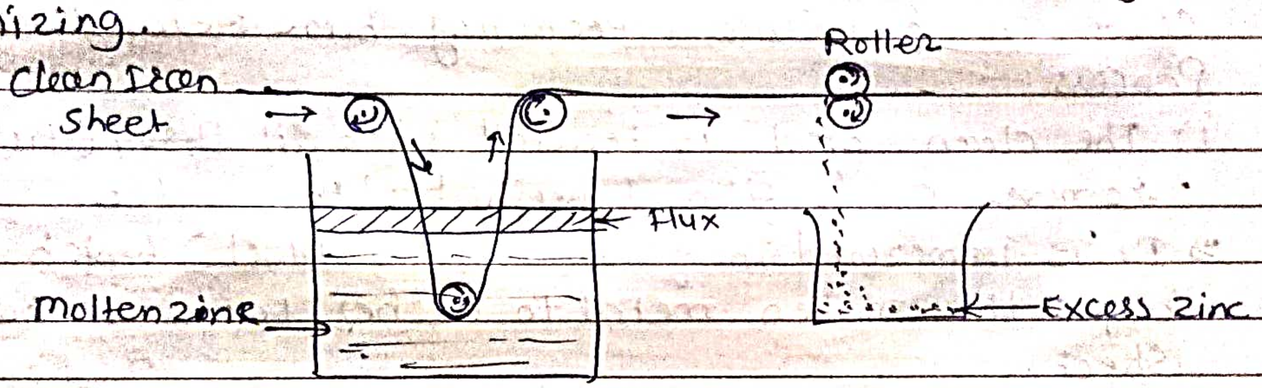

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4) The metal article is then dipped in bath of molten Zn at 425°c to 430°c
5) The surface of bath is covered with flux ($NH_4Cl + ZnCl_2$) Flux cleans the metal surface before coating for better adhesion & prevent the oxidation.
6) It is then passed thr pair of hot roller to

remove excess zinc & produce thin uniform coating.

7) It is then annealated at 650°c & cooled to room temp. slowly.

Appl^n :-

wires, pipes, buckets, tubes, screw, nails etc.

Limitations :-

1) Can not used to storing foods as it formed poisonous product by the action of food on zinc.

2) If coating get rupture then corrosion takes place faster.

b) Tinning :- (tin)

" Coating of tin on steel, is known as tinning.
~~This~~ Tin being more cathodic, give better corrosion resistance, protecting base metal (Iron)

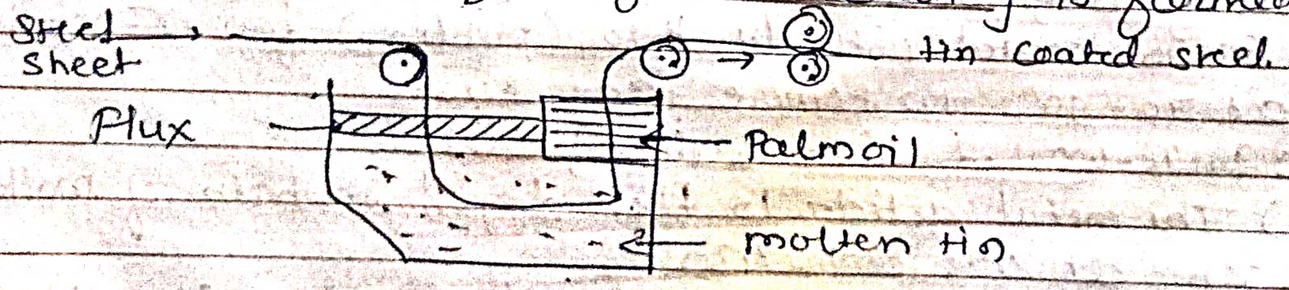
Process :-

1) The clean steel is treated with dil H_2SO_4 (Pickling) remove oxide film & washed with water.

2) It is immersed in flux ($ZnCl_2$ & NH_4Cl) which ~~remove~~ helps the molten metal to adher to the metal sheet.

3) The metal article is then passed thr a tank of molten tin at temp. 240°c

4) Finally passed thr a series of roller which remove excess of tin & uniformed coating is formed.



Uses :- Used in stooring food, ghee, oil, pickels. medicine etc.

2) Tinned copper vessels used for cooking utensils. etc.

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Galvanising	Tinning.
1) Zn coating on iron	1) Tin coating on iron.
2) Temp. for coating is about 450°C	2) Temp for coating is about 250°C
3) Anodic coating	3) Cathodic coating
4) Applicable for general use like sheets, pipes, wires, angles	4) Applicable for containers for storing edible material.
5) Reliable, Zn is placed higher in gavanic series	5) Less reliable, lower placed in galvanic series
6) Cheaper	6) Costlier.

* Electroplating :-

In this method coating metal is coated on a base metal on the basis of electrolysis process.

Objectives -

i) To increase corrosion resistance

ii) to get better appearance

iii) To get increased hardness

iv) To change the surface properties of metal & non metal.

Process :-

1) Base metal is cleaned to remove oils, greases, oxides

2) metal articles is made as cathode & coating metal is anode

3) No. of cathod & Anode suspended for uniformed coating.

4) Cathode & Anode dipped in sol.ⁿ i.e electrolyte (salt sol.ⁿ of coating metal)

5) Suitable current is applied.

6) The anode metal get oxidised & enter the electrolyte sol.ⁿ. Cathode get reduced & dipoites on the surface of base metal.

(2) At cathode eg -

$$Ag^{+} + e^{-} \longrightarrow Ag \quad (\text{If Ag plating})$$

(1) At Anode . $Ag \longrightarrow Ag^{+} + e^{-}$ (If Ag anode)

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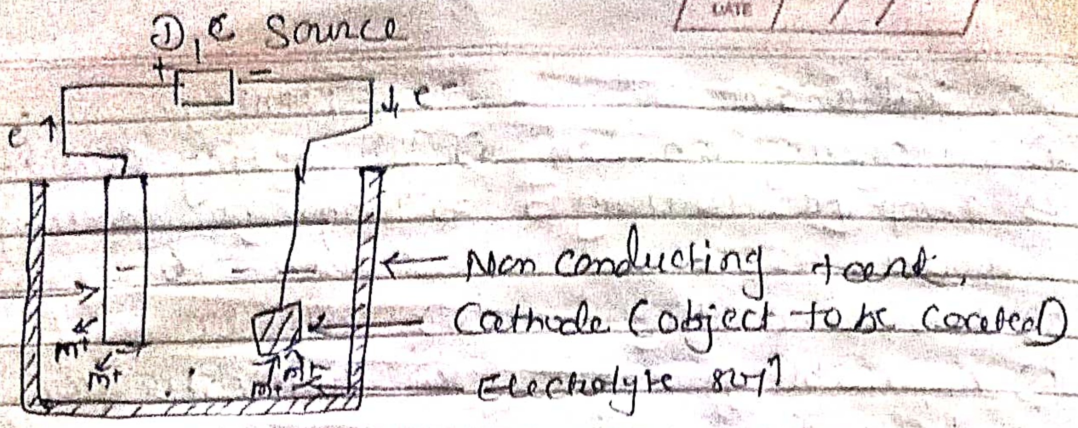

Electroplating.

* Factors affecting electroplating process
1) Control of current density
2) Control of temp.
3) Control of conc. of ion & pH.
4) Surface area of anode & position.

*Appl?
i) corrosion protection eg. Chromium plating on steel.
2) decorative or better appearance. - gold, Ag, plated steel, used in ornaments, watches.
3) In Automobile, aircraft, radio industries used thick layer of metal by electroplating.
4) Non metallic material like wood, paper, glass electroplated for decoration, preservation of surface.
eg. synthetic resin are electroplated with copper.

* Advantages:-
1) Coating thickness can be control.
2) non expensive, i.e less cost
3) fine coating can be obtained.

* Metal Cladding :-
" Is the process in which thin sheet of coating metal is firmly bonded with mechanical pressure to the base metal on both side.
- the coating metal is called cladding metal.
- metal like Ni, Cu, Ag, Pt. & alloys are used for metal cladding.
- Base metal on which cladding done are like

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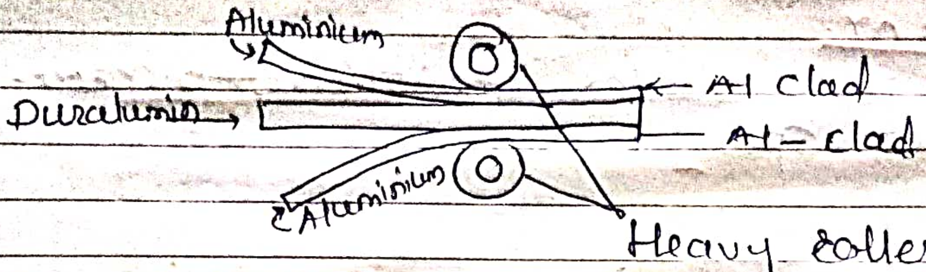
mild steel, Cu etc or alloys.

- Cladding is done by performing a sandwich of layer of base metal, which are passed thr rollers & bonded under the action of heat & pressure.

eg - Aluminium clad sht used in aircraf industry
Cu clad used in electrical industry.



* Cementation / Diffusion coating *

* Cementation :-

formation of a strong layer of alloy of coating metal & base metal on the surface of base metal.

- It is obtained by heating base metal in revolving drum containing a powder of the coating metal.
- The powder coating metal diffuses into the surface of the base metal.
- formation of alloy of coating metal.

eg - coating of small articles like bolts, screws, valve gauge tools etc

a) Sherardizing

If coating metal is zinc.

- Iron articles are cleaned & packed with Zn dust in a drum & rotated for 2-3 hr At temp. 350-370°C. Zn diffuse into Fe & formed Fe-Zn alloy surface.

b) Colorizing :-

coating of metal is Aluminium.

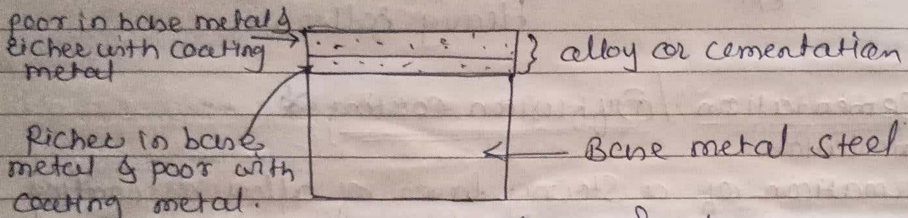
- metal objects are heated in drum containing Al powder & Al-oxide. heated at 4 to 6 hr at 840°C to 930°C.

This method is used for protection of furnace parts, valves, condenser in oil refineries.

c) Chromizing -

If coating metal is chromium.

- Heating base metal in drum which contain Chromium powder & Alumina at 1300 to 1400°C at for 3-4 hr.
- Used in turbine blades, resistance to corrosion by salt spray.



cementation surface